

ITU-MiniTwit — BSc DevOps, Software Evolution and Software Maintenance

Group: BSc_group_m

Repository: <https://github.com/RonoITU/itu-devops2026-jackhammers>

Issue tracker: <https://github.com/RonoITU/itu-devops2026-jackhammers/issues>

Monitoring dashboard: <http://178.104.27.224:3000/d/chirp-aspnet-001/windysquirrels-monitoring-d>

Logging dashboard: <http://178.104.27.224:3000/d/app-logging-dashboard/windysquirrels-logging-d>

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1. System's Perspective

1.1 Design and Architecture

Author(s): Rasmus Alexander Christiansen

This section describes the overall system architecture of MiniTwit, including how the application is structured, deployed, and monitored. The diagram above provides a visual overview of the components and their interactions across the infrastructure.

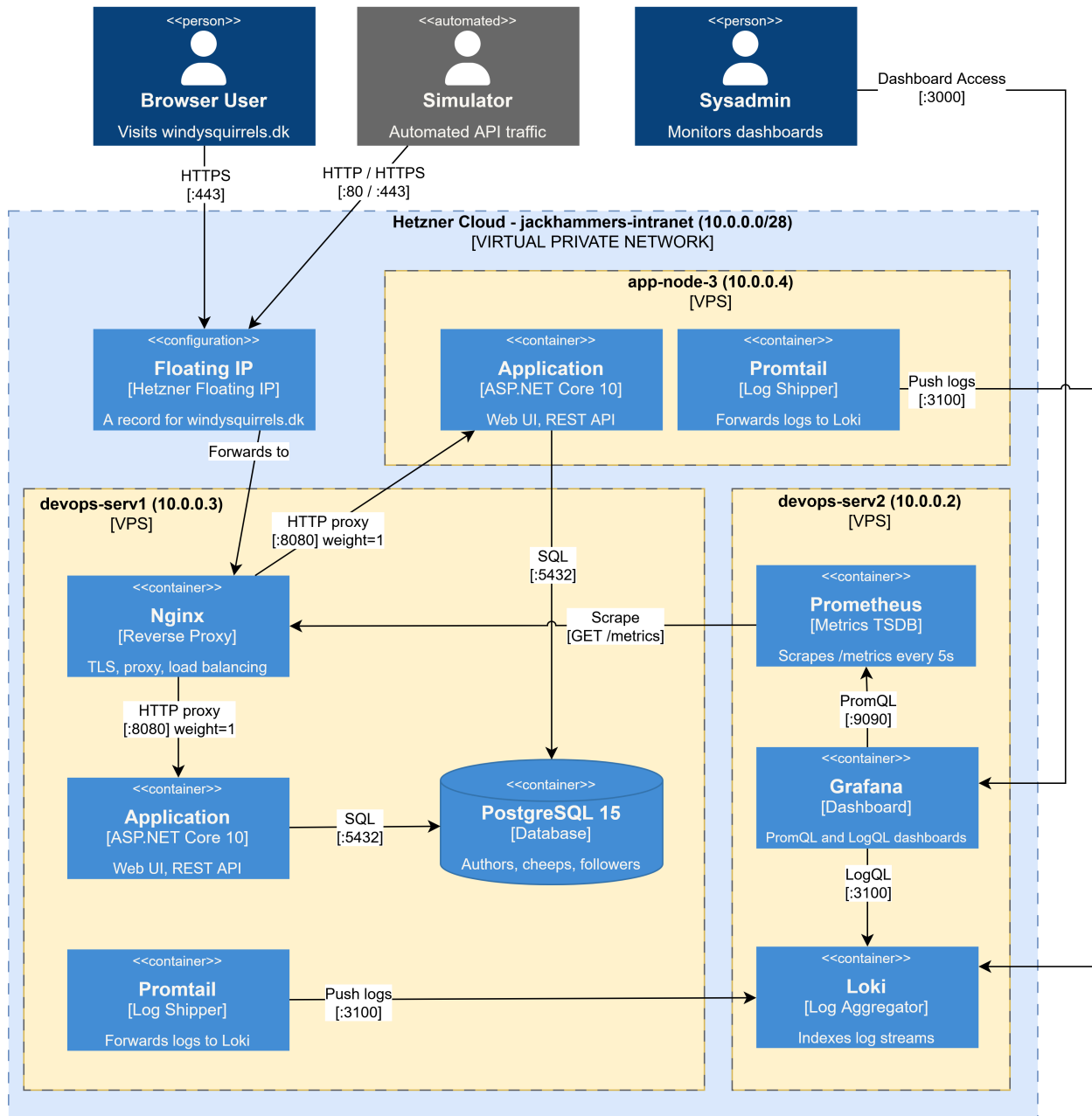


Figure 1: Diagram of the system architecture

MiniTwit is deployed across three Hetzner Cloud VPS nodes connected via a virtual private network

(10.0.0.0/28).

devops-serv1 (10.0.0.3) is the primary node. Nginx runs here as the sole public entry point, handling TLS termination via Let's Encrypt, redirects HTTP traffic to HTTPS, and weighted load balancing across both app nodes. The primary application instance serves web UI and REST API traffic and exposes a /metrics endpoint for Prometheus. PostgreSQL 15 hosts the shared database for all application data. Promtail runs as a log shipping agent, collecting Docker container logs from this node and forwarding them to Loki on devops-serv2.

app-node-3 (10.0.0.4) is the secondary application node. It runs an application replica that receives the majority of traffic from Nginx (weight=2) and connects to the shared PostgreSQL database on devops-serv1. Promtail runs here as well, shipping container logs to Loki on devops-serv2.

devops-serv2 (10.0.0.2) is dedicated to observability. Prometheus scrapes the /metrics endpoint on devops-serv1 every five seconds and stores the resulting time-series data. Loki aggregates the structured log streams pushed by the Promtail agents on devops-serv1 and app-node-3. Grafana provides dashboards over both data sources using PromQL for metrics and LogQL for logs.

1.2 Dependencies

Author(s): Rasmus Alexander Christiansen

The diagram below illustrates the relationships between the key technologies across all layers of the system. Not every dependency is shown - minor libraries and transitive dependencies are omitted for clarity. A full list with descriptions is provided in the tables below the diagram.

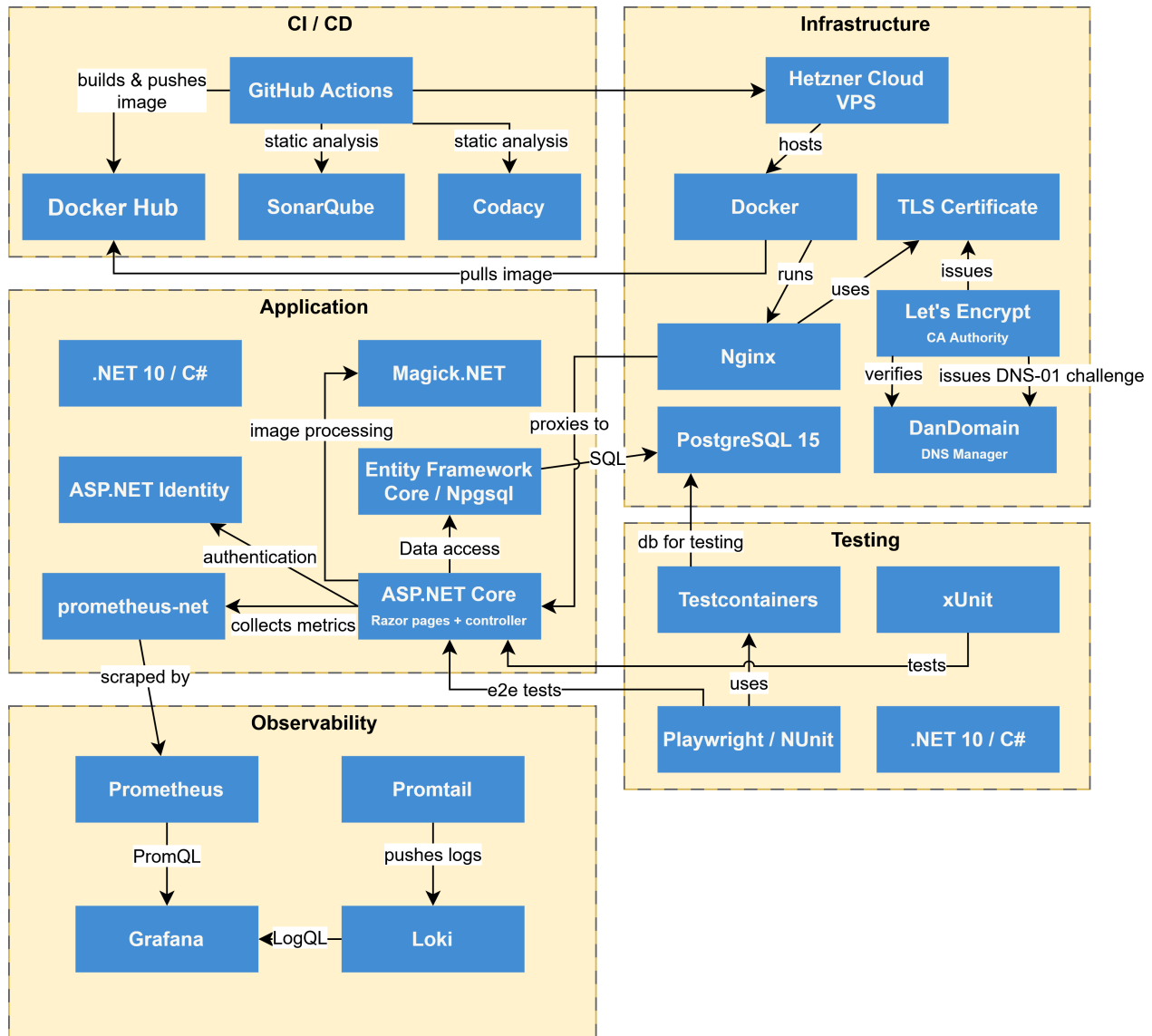


Figure 2: Diagram of the system dependencies

Application

Technology / Tool	Purpose
C# / .NET	Application language and runtime
ASP.NET Core (Razor Pages)	Web UI framework
ASP.NET Core (Controllers)	REST API for the simulator
Entity Framework Core	ORM and database migrations
Npgsql EF Core Provider	PostgreSQL driver for EF Core
ASP.NET Core Identity	User authentication and account management
prometheus-net	Exposes application metrics to Prometheus
Magick.NET	Profile image processing
SixLabors.ImageSharp	Image resizing and encoding

Infrastructure

Technology / Tool	Purpose
Hetzner Cloud	VPS hosting for all three nodes
Docker	Container runtime on all nodes
Docker Compose	Service orchestration on each node
Nginx	Reverse proxy, TLS termination, load balancing
PostgreSQL	Relational database
Let's Encrypt / Certbot	Automated TLS certificate provisioning

Observability

Technology / Tool	Purpose
Prometheus	Metrics scraping and time-series storage
Grafana	Dashboards for metrics and logs
Loki	Log aggregation and indexing
Promtail	Log shipping agent (Docker container logs)

CI/CD

Technology / Tool	Purpose
GitHub Actions	CI/CD pipeline automation
Docker Hub	Container image registry
SonarQube (SonarCloud)	Static analysis and quality gate
Codacy	Additional static analysis
Pandoc + XeLaTeX	Report PDF generation

Testing

Technology / Tool	Purpose
xUnit	Unit and integration test framework
NUnit + Playwright	End-to-end browser testing
Testcontainers	Spins up a PostgreSQL instance for tests
coverlet	Code coverage collection

1.3 Current State of the System

Author(s): Rasmus Alexander Christiansen

The current state of the system is assessed through two static analysis tools integrated into the CI pipeline: SonarCloud and Codacy. Both run automatically on every pull request. Rather than blocking merges on quality gate failure (**continue-on-error: true** in the pipeline), the team has deliberately prioritised visibility over enforcement - the goal being that issues are made transparent and trackable, not that velocity is sacrificed to a strict gate.

SonarCloud

SonarCloud provides a continuous quality assessment of the codebase across security, reliability, maintainability, coverage, and duplications.



Figure 3: SonarCloud project summary

It is worth noting that the project was inherited from a 1.5 year old codebase originally written by students in the third-semester BDSA course. As a result, a significant portion of the issues identified by SonarCloud reflect legacy code rather than code introduced during this course. The team has primarily focused on resolving security issues and issues affecting performance, while reliability, maintainability, and code duplication issues in the inherited code have not been a priority.

The overall results show 1 security issue, 41 reliability issues, 106 maintainability issues, 75.4% test coverage, and 8.7% code duplication. There are 0 security hotspots and all hotspots have been reviewed.

The Issues view illustrates what this visibility looks like in practice:

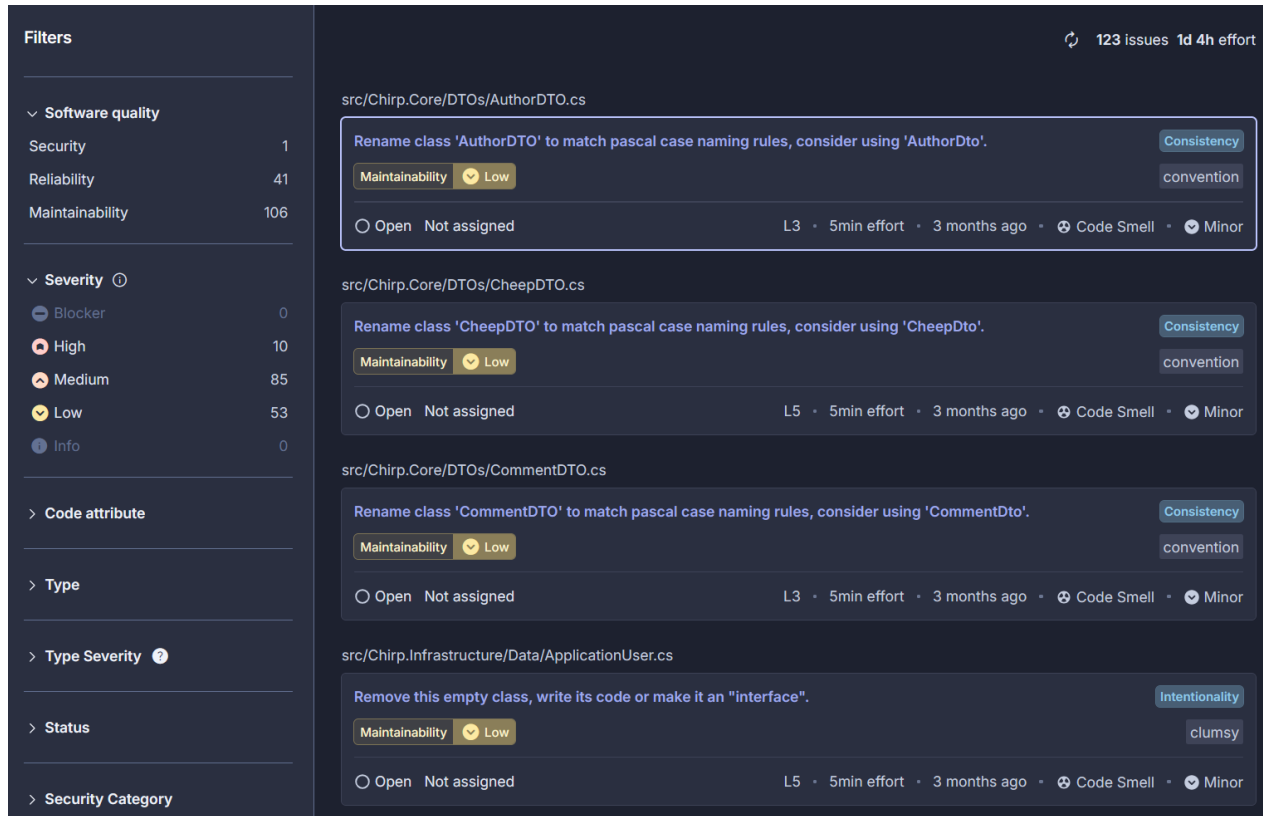


Figure 4: SonarCloud issues

SonarCloud surfaces concrete, prioritised issues directly tied to source locations - for example, flagging that `AuthorDTO` in `src/Chirp.Core/DTOs/AuthorDTO.cs` should be renamed to `AuthorDto` to conform to Pascal case naming conventions. Before static analysis was introduced, issues like this existed in the codebase but were invisible. They are now explicit, categorised by severity, and can be worked through systematically.

Codacy

Codacy runs as a secondary static analysis tool in the CI pipeline alongside SonarCloud, providing an independent second opinion on code quality. It does not currently expose a public dashboard, but its results are available in the pipeline logs on every run.

2. Process' Perspective

2.1 CI/CD Pipeline

Author(s):

2.2 Monitoring

Author(s):

2.3 Logging

Author(s):

2.4 Security Hardening

Author(s):

2.5 Availability and Scaling

Author(s):

3. Reflection Perspective

3.1 Evolution and Refactoring

Author(s):

3.2 Operation

Author(s):

3.3 Maintenance

Author(s):

3.4 DevOps Reflection

Author(s):

4. Use of Generative AI

Author(s): Rasmus

GitHub Copilot was used to assist with boilerplate code generation throughout development. It provided inline auto-completion that often matched the intended behavior being implemented, and helped untangle unfamiliar functionality across the broader codebase. This sped up the coding process significantly, particularly for repetitive or structurally predictable code.

Microsoft Copilot was used for general consultation when extending the infrastructure, serving as a quick reference for exploring options and approaches before committing to a direction.

Claude was used for more in-depth technical discussions, typically involving specific code examples. It helped provide clarity on complex topics and was particularly useful when a concept required more thorough explanation than a quick search could offer.